Truth, Lies, and Reasoning Machines

How Language Models Handle Misinformation and Self-Verification

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Abstract

Large language models (LLMs) are often deployed in environments where input information may be incomplete, misleading, or factually incorrect. Understanding how these systems respond to misinformation is therefore important for evaluating their reliability. In this project, we investigate how LLMs behave when exposed to misleading claims and whether explicit self-verification prompts improve factual accuracy.

We evaluate two models, Claude and Qwen-2.5-7B-Instruct, using a fixed subset of 50 questions from the TruthfulQA (Lin, Hilton, & Evans, 2022) dataset. Each question is presented under three controlled prompting conditions: a baseline condition with no misleading context, a noisy condition where an incorrect claim is explicitly marked as unreliable, and an adversarial condition where an incorrect claim is framed as authoritative or widely accepted. For each condition, the model produces an initial answer and a second answer after a self-verification prompt.

We evaluate factual accuracy before and after verification, along with consistency related metrics including self-correction and regression. Our results show that Claude maintains consistently high accuracy across conditions, while Qwen exhibits lower initial accuracy but benefits more from explicit self-verification, particularly under adversarial framing. Overall, the findings suggest that verification prompts can improve robustness to misinformation and that authority-based framing does not always increase error rates but may instead trigger more cautious reasoning.

Introduction

LLMs are increasingly used as general purpose systems for answering questions, summarizing information, and assisting with decision making. In real world applications, however, the information provided to these models is not always reliable. Prompts may contain incorrect assumptions, incomplete knowledge, or statements presented as authoritative even when they are factually wrong. Understanding how language models behave under such conditions is therefore essential for evaluating their reliability and limitations.

One common failure mode of language models is the uncritical acceptance of false premises. When a prompt includes an incorrect assumption, the model may treat that assumption as true and generate an answer that is logically consistent with the prompt but factually incorrect. Recent work has also shown that factual knowledge inside language models can be localized and manipulated within model representations, highlighting the structured nature of model knowledge (Meng, Bau, Andonian, & Belinkov, 2022). Another concern is sensitivity to authority-based framing. Phrases such as "many experts say" or "it is widely believed that" may influence the model's reasoning even when the claim itself is false (Zhou, Wang, Wang, Tang, & Liu, 2023).

In this project, we study how language models respond to misleading information and whether explicit self-verification prompts can improve factual reliability. We use the TruthfulQA dataset, which is specifically designed to test whether models reproduce common misconceptions or provide factually correct answers.

From this dataset, we select a fixed subset of 50 questions and evaluate model responses under three different prompting conditions: baseline, noisy, and adversarial. In addition, we introduce a self-verification step in which the model is asked to review and verify its own answer. This allows us to study both the effects of misleading context and the potential benefits of introspective prompting.

To explore these questions, we evaluate two models with different characteristics: Claude, a large proprietary model accessed via API, and Qwen-2.5-7B-Instruct, a smaller open-weight instruction-tuned model accessed through the Hugging Face inference API. By comparing their behavior across conditions, we aim to better understand how misinformation and verification prompts influence LLMs reliability.

Methods

1. Dataset

We base our experiments on the TruthfulQA dataset, which contains questions designed to extract common misconceptions alongside reference correct answers and

known incorrect answers. The incorrect answers provided by the dataset are particularly useful for constructing prompts that contain realistic but false information.

To ensure consistency across models and conditions, we select a fixed subset of 50 questions. The same subset is used for all experiments, allowing for direct comparison between models and prompt conditions.

2. Prompt Conditions

Each question is evaluated under three controlled prompting conditions:

- **Baseline**

  The model receives only the question, with no additional context. This condition measures standard question-answering performance.

- **Noisy**

  The question is accompanied by an incorrect claim that is explicitly presented as potentially wrong. This condition tests whether the model can ignore unreliable information when it is clearly marked as such.

- **Adversarial**

  The question is accompanied by an incorrect claim framed as widely accepted or authoritative (for example, “many experts say”). This condition tests whether social or authority-based framing increases the likelihood that the model will accept false information.

In both the noisy and adversarial conditions, the incorrect claim is taken directly from the set of incorrect answers provided by TruthfulQA.

3. Self-Verification Prompt

To evaluate introspective reasoning, each response is generated in two stages. First, the model produces an initial answer under one of the three conditions. Second, the model is shown its own answer and prompted to verify every factual claim, correcting any unsupported or incorrect statements. The model is also instructed to explicitly indicate whether changes were made.

This verification step allows us to measure whether explicit self-checking improves factual accuracy and reasoning stability.

4. Models

We evaluate two language models:

- **Claude**,
  accessed via the Anthropic API, representing a large, closed, high-capacity model.
- **Qwen-2.5-7B-Instruct**,
  accessed via the Hugging Face Inference API, representing a smaller open-weight instruction-tuned model.

Both models are evaluated using the same dataset subset, prompt templates, and evaluation pipeline.

5. Pipeline

The experimental workflow follows a reproducible pipeline:

Dataset subset (50 questions)

→ prompt generation (baseline, noisy, adversarial)

→ model inference

→ raw output collection

→ manual annotation of responses

→ metric computation

→ visualization and analysis

Raw model outputs are stored as JSONL files, while annotation tables record whether the first and verified answers are factually correct.

Results

1. Hallucination Behavior Across Prompt Conditions

We first analyze how often models produce hallucinated or factually unsupported answers under different prompt conditions. Figure 1 shows the hallucination rate for Claude and Qwen across the baseline, noisy, and adversarial settings.

Claude maintains relatively low hallucination rates across all conditions. In the baseline setting, the hallucination rate is approximately 10%. This decreases slightly in adversarial conditions to around 6%, suggesting that the presence of strong contextual framing does not necessarily increase hallucination behavior.

Qwen shows slightly higher hallucination rates overall. In the baseline condition, the hallucination rate is approximately 12%, while both noisy and adversarial conditions produce rates around 8% and 4%, respectively.

Interestingly, adversarial framing does not increase hallucinations for either model. Instead, it appears to encourage more cautious reasoning, particularly for Qwen, where hallucinations decrease noticeably. This suggests that strong contextual signals may sometimes trigger more careful evaluation rather than blind acceptance of misleading information.

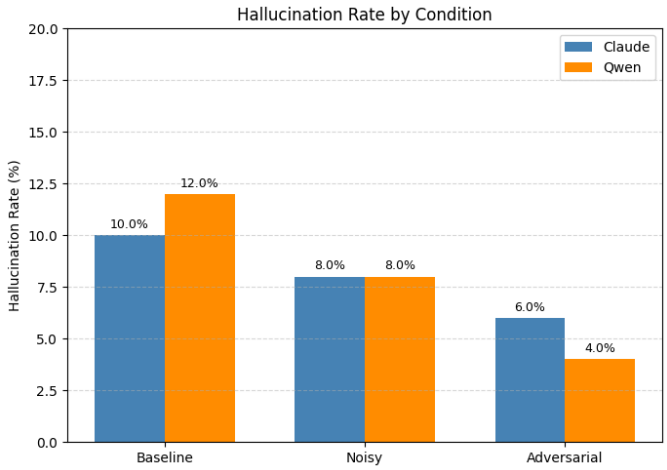

Figure 1: Hallucination rate across prompt conditions for Claude and Qwen

2. Effect of Self-Verification

To evaluate the effect of the verification prompt, we analyze how often models correct their own mistakes or introduce new errors during the verification step.

Figure 2 shows the number of cases where verification changes the correctness of an answer. Two behaviors are measured:

- **Self-correction:** an incorrect answer becomes correct after verification
- **Regression:** a correct answer becomes incorrect after verification

Claude shows relatively stable behavior, with only a few cases of both self-correction and regression. Qwen, on the other hand, exhibits a larger number of self-corrections, indicating that verification prompts help it fix a greater proportion of initial mistakes. However, Qwen also shows slightly higher regression rates, meaning that verification occasionally destabilizes correct answers.

Overall, these results suggest that verification prompts provide greater benefits for weaker models, while stronger models already maintain relatively stable reasoning.

Figure 2: Self-correction and regression after verification

3. Accuracy Improvement After Verification

Claude achieves higher accuracy than Qwen in all conditions. In the baseline condition, Claude reaches **84% accuracy**, while Qwen achieves **76%**. In the noisy conditions, Claude's accuracy increases slightly to **88%**, while Qwen drops to **74%**. In adversarial conditions, Claude reaches **90%**, while Qwen returns to **76%**.

Overall, the results show that Claude maintains consistently higher first-answer accuracy across conditions. The noisy prompts appear to have a small negative effect on Qwen, while Claude remains relatively stable. These results suggest that the two models respond differently to variations in prompt framing.

Figure 3 shows the accuracy of the models' **first responses** under the three prompt conditions: baseline, noisy, and adversarial.

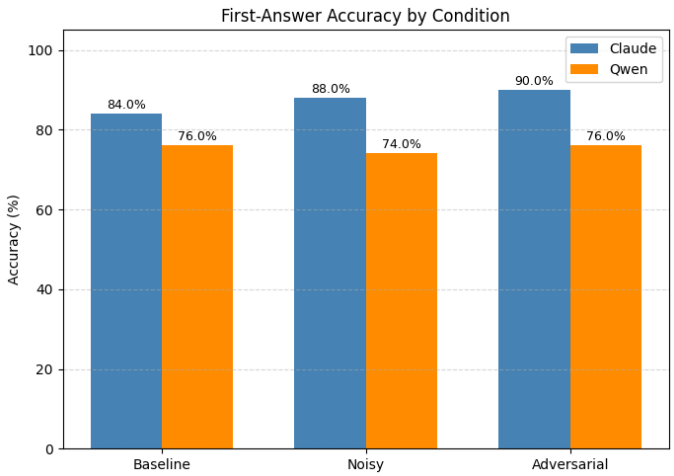

Figure 3: First-answer accuracy (%) across prompt conditions.

4. Performance Across Knowledge Categories

To better understand where models struggle, we analyze performance across different categories of questions in the TruthfulQA dataset.

Figure 4 shows the first-answer accuracy across categories for both models. Some domains, such as language, proverbs, and logical falsehoods, are solved reliably by both models. Other domains, such as advertising, superstitions, and economics, show greater variation in accuracy.

Qwen generally performs worse than Claude across most categories, although both models exhibit similar patterns of difficulty across topics.

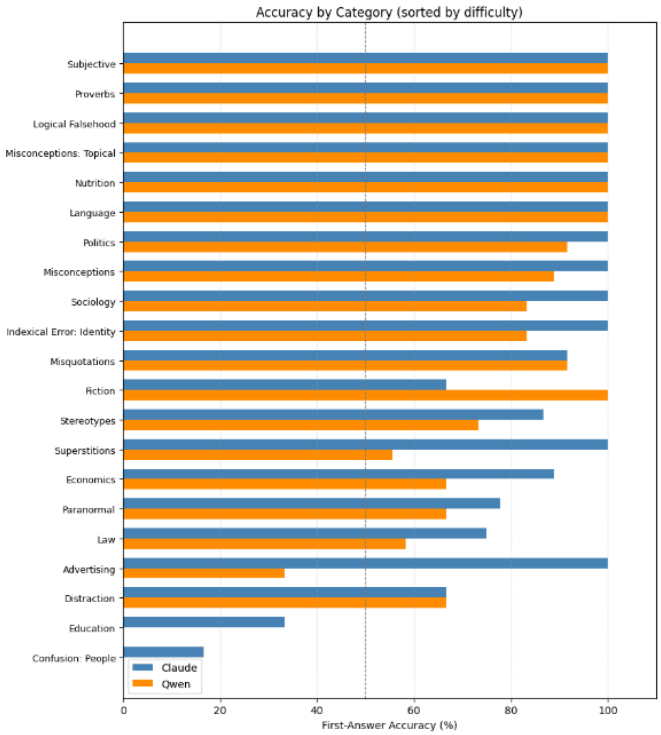

Figure 4: Accuracy by category for Claude and Qwen

Summary of Findings

Overall, the results show that:

- Claude is robust in all conditions, with high accuracy and low regression.
- Qwen has lower initial accuracy but gains more from explicit self-verification in the adversarial framing.
- Adversarial framing does not necessarily increase error and can sometimes encourage more careful reasoning.
- Many incorrect responses are due to reliance on misleading premises rather than hallucination.

These findings highlight the importance of both prompt design and self-verification when evaluating and improving LLM reliability under misinformation.

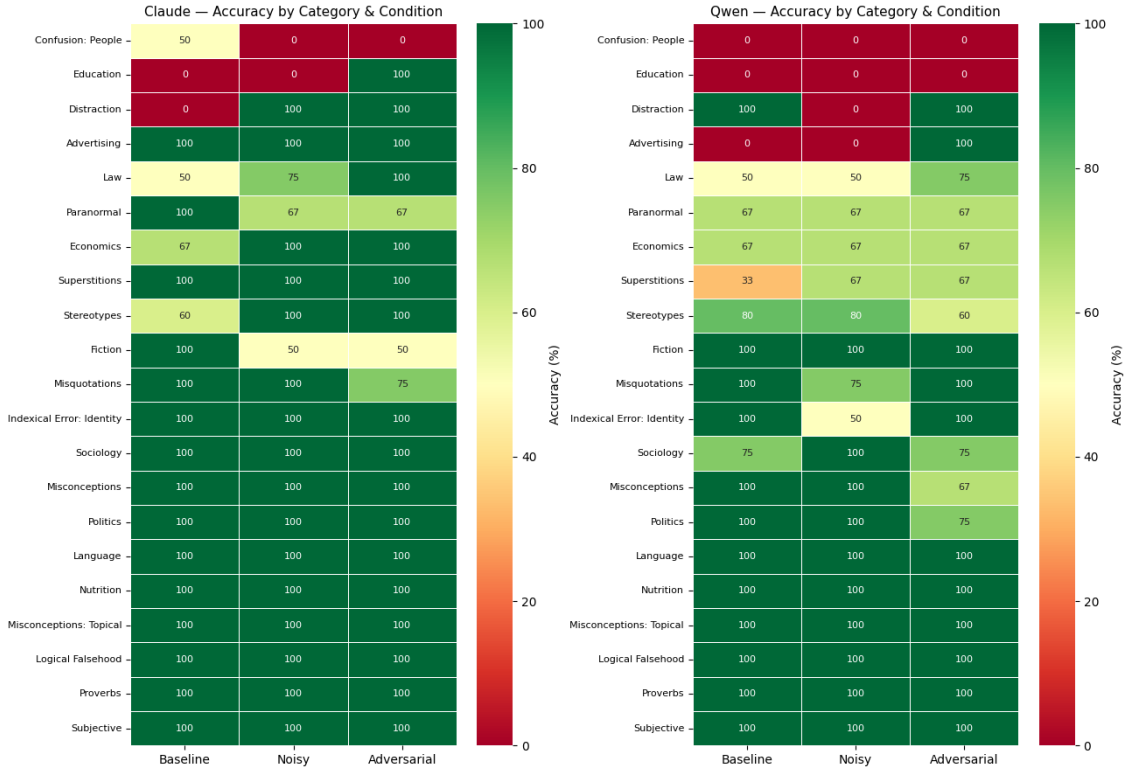

Figure 5: Accuracy heatmap across categories and prompt conditions

Conclusion

In this project, we examined how language models respond to false, noisy, and adversarial information, and whether explicit self-verification prompts can improve reliability. The experimental results show that large models such as Claude are generally robust across all conditions, while smaller models such as Qwen-2.5-7B-Instruct exhibit lower initial accuracy but benefit more from explicit verification, particularly when misinformation is framed as authoritative.

A key finding is that adversarial framing does not necessarily increase error rates. In several cases, especially for Claude, authoritative but incorrect claims trigger more cautious reasoning rather than blind acceptance. This suggests that language models may sometimes treat strong contextual cues as signals to engage in more deliberate evaluation rather than as facts to be repeated. At the same time, qualitative analysis shows that rejecting a false premise does not always lead to recovery of the correct answer, highlighting a distinction between epistemic caution and full factual correctness.

Self-verification emerges as a useful but imperfect mechanism. While it enables both models to correct a subset of their initial mistakes, it can also introduce regressions, particularly for smaller models. This indicates that introspective prompting improves robustness but may also increase instability when the model's underlying knowledge is weak.

Several directions for future work emerge from this study. First, evaluating a larger and more diverse subset of TruthfulQA would improve statistical reliability. Second, extending

the analysis to additional models and architecture could clarify how robust scales with model size and training. Third, combining behavioral analysis with internal explainability methods, such as token attribution or confidence estimation, could provide deeper insight into the mechanisms behind these behaviors. Finally, future work could explore alternative verification strategies that reduce regression while preserving the benefits of self-correction.

AI Usage Disclaimer

Parts of this project were developed with the assistance of generative AI tools, including **OpenAI's ChatGPT (GPT-5)** and **Google AI Studio**. These tools were used to support the development of project ideas, discuss and refine the experimental design, assist with code development, and help draft and edit descriptive sections of the report.

All AI-assisted outputs were carefully reviewed, modified, and validated by me. I take full responsibility for the final content, including the structure of the project, the implementation of experiments, the interpretation of results, and the academic integrity of the work.

All conclusions, evaluations, and methodological choices reflect my own reasoning and judgment.

References

Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods.

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